# ECE LabMate

AI-Powered Interactive Learning for Electronics and Communication Engineering | Gnani AI Internship Challenge

## The Problem

ECE students often learn complex concepts through theory alone, while practical visualization and regular viva practice may be limited. This can make topics such as signals and modulation harder to understand and apply.

## The Solution

**ECE LabMate** is an interactive learning application that brings AI tutoring, voice-based explanations, virtual experiments, and viva self-assessment together in one MATLAB App Designer interface. It aims to make ECE learning more accessible, engaging, and practice-oriented.

## Key Features

- **AI ECE Tutor:** Uses Gemini AI to provide student-friendly explanations of ECE concepts.
- **Voice Learning:** Integrates Gnani AI Text-to-Speech to convert responses into audio.
- **Virtual Lab:** Enables students to generate and visualize signals and explore modulation experiments.
- **Viva Practice:** Presents practice questions and evaluates responses using keyword matching with a score for self-assessment.

## Technology

MATLAB App Designer • Gemini AI API • Gnani AI TTS API • Signal Processing and Communication Systems concepts

## Value and Impact

ECE LabMate connects theoretical learning with interactive practice. By allowing students to ask questions, listen to explanations, visualize experiments, and test their understanding, it supports independent learning and helps students identify topics that need more revision. The project demonstrates practical integration of AI services into an engineering learning application.

## Project Demo and Source

GitHub Repository: https://github.com/KesamreddyArpitha/ECE-LabMate

Demo video: Available through the repository README / linked YouTube demo.